



Skills for AI Agent Workshop

- AI Agents Instruction
- Why Skills
- Skill Architecture & Progressive Disclosure
- Skill in Agentic Ecosystem
- Building the Skill
- Skill Deployment, Invocation and Management
- Hands-on Workshop

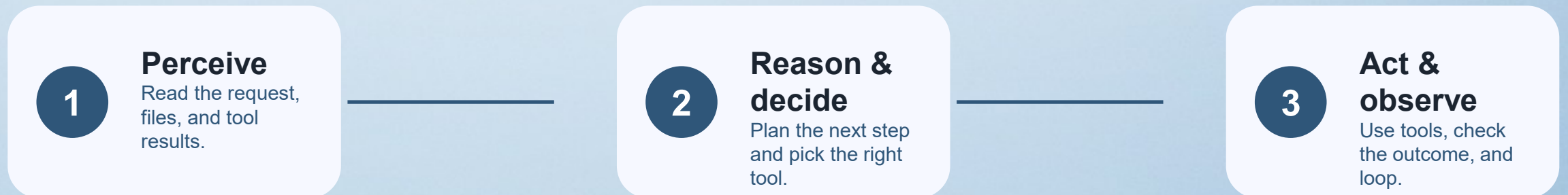
AI Agents Instruction



What is an AI agent?

An AI agent is an LLM-powered system that pursues a goal on our behalf.

It perceives its environment, reasons about the next step, and takes action with tools — then observes the results and keeps looping until the task is done.



Effective agents combine three ingredients

1

A capable model

Reasoning and language ability at the core to understand the goal and plan the work.

- OpenAI
- Anthropic
- Gemini

2

Tools & data access

Search, code, files, and APIs let the agent act on the world, not just talk about it.

- Read
- Edit
- Run
- Write

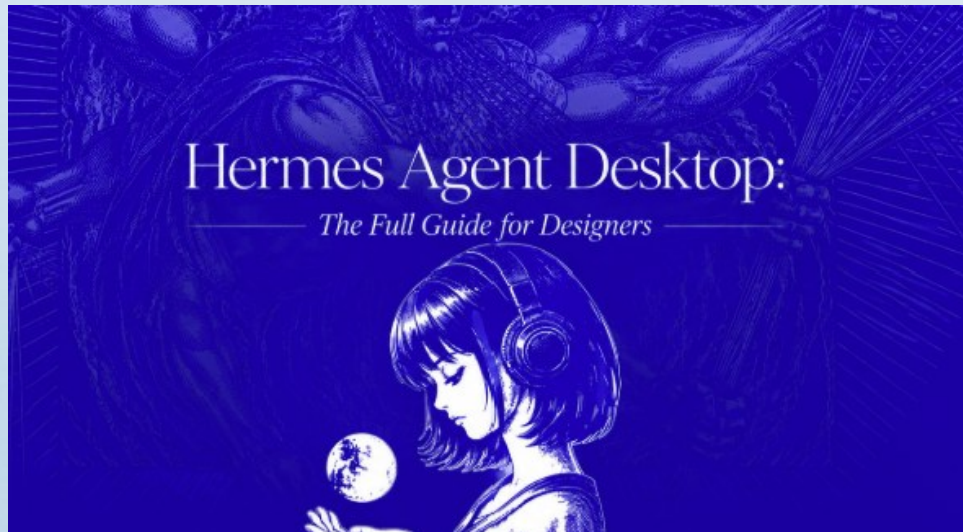
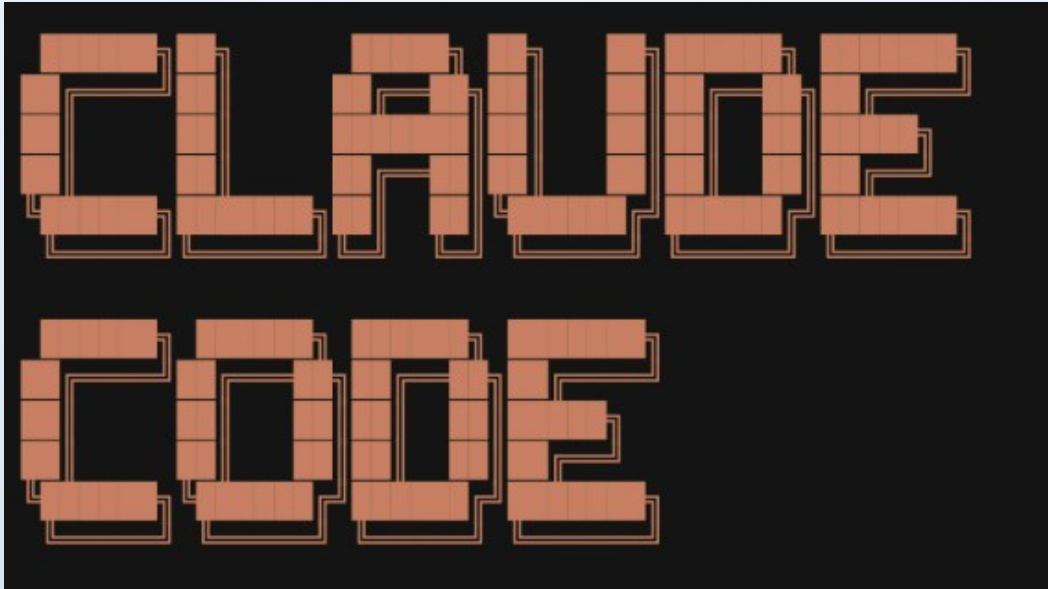
3

Reusable expertise

Skills bundle instructions, references, and scripts so every run follows the same method.

This workshop focuses on the third ingredient — turning your team's know-how into skills an agent can load on demand.

AI Agents





Uploading “kung fu” to Neo



I know kung fu.



Uploading “flying the helicopter” skill to Trinity

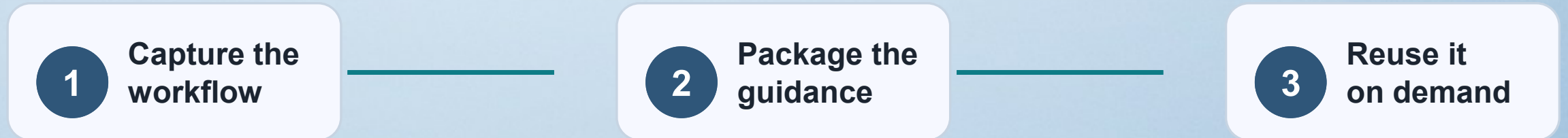


Why Skills?



Skills make good work repeatable

Turn recurring instructions into a capability an agent can apply consistently.



Reusable skills replace repeated prompting

Without a skill

Restate the instructions



Reattach the rules

Explain the output



Correct inconsistencies

With a skill

1

Upload once

Instructions and references stay together.

2

Invoke when needed

The same method drives each run.

Skill Architecture & Progressive Discloser



Agent skills are discoverable capability folders

II Why agent skills

skill-name/

```
SKILL.md  
scripts/  
assets/  
references/
```

A lightweight, open format for extending what an AI agent can do.

- Organizes instructions, scripts, assets, and resources.
- Lets agents discover the right capability for a specific task.
- Improves accuracy by packaging the method with its supporting context.

The specification keeps the core simple

Required

SKILL.md

Defines the skill, when to use it, and the workflow to follow.

references/

Detailed guidance and domain knowledge

scripts/

Executable helpers and repeatable checks

assets/

Templates, images, and output resources

Optional folders are added only when the task needs them.

Real skills can package very different resources

pdf/

```
|— SKILL.md
|— references/.md
|   |— forms.md
|   |— reference.md
└— scripts/
    |— check_fillable_fields.py
    |— convert_pdf_to_images.py
    |— extract_form_field_info.py
    └— fill_pdf_form_with_annotations.py
```

designing-newsletters/

```
|— SKILL.md
|— references/
|   └— style-guide.md
└— assets/
    |— header.png
    |— icons/
    └— templates/
        |— newsletter.html
        └— layout.docx
```

General-purpose agents replace specialized stacks

Old paradigm

Research agent

Coding agent

Finance agent

Marketing agent

New paradigm

General-purpose agent

Code is the universal interface



Skills add the context and domain expertise needed for reliable work.

- **Research skill**
- **Coding skill**
- **Finance skill**
- **Marketing skill**

Skills deliver three kinds of on-demand context

1

Domain expertise

Brand guidelines and templates
Legal review processes
Data analysis methods

2

Repeatable workflows

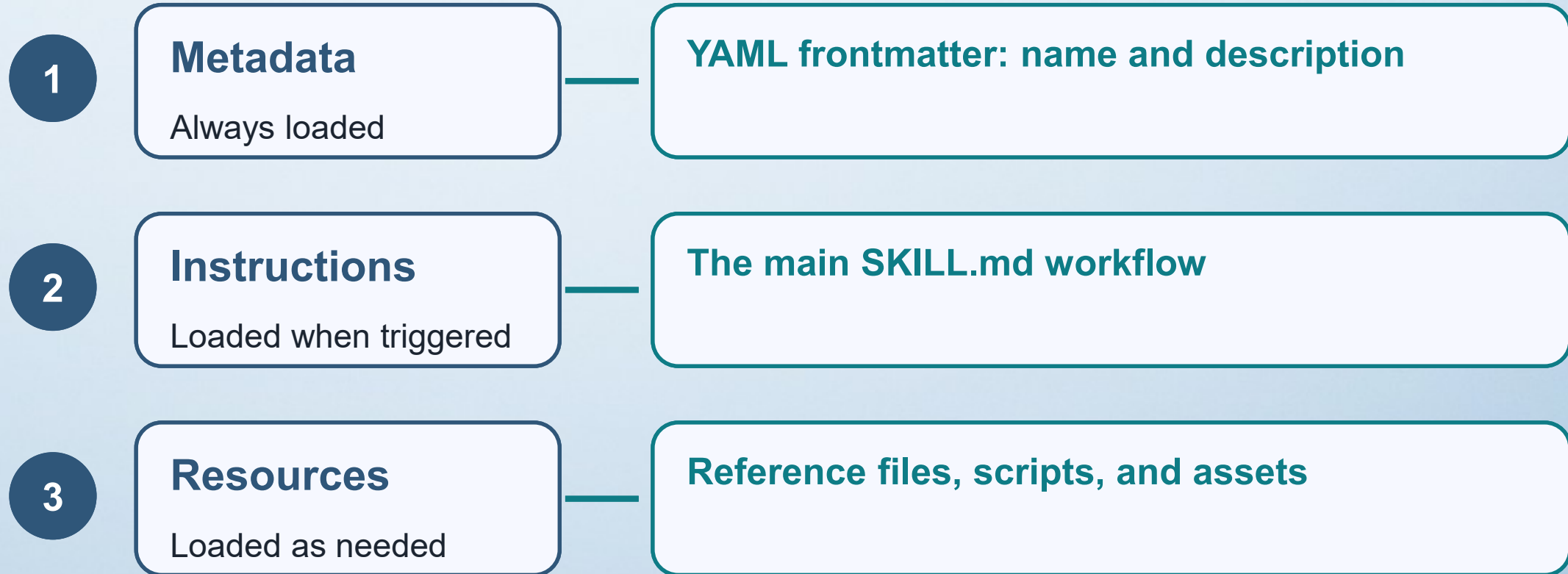
Weekly campaign review
Customer call preparation
Quarterly business review

3

New capabilities

Create presentations
Generate Excel or PDF reports
Build MCP servers

Progressive disclosure protects the context window



Hundreds of skills can exist without loading all their details at once.

Progressive disclosure – 1. metadata



pdf

```
---
name: pdf
description: Use this skill whenever the user wants to do
anything with PDF files. This includes reading or
extracting text/tables from PDFs, combining or merging
multiple PDFs into one, splitting PDFs apart, rotating
pages, adding watermarks, creating new PDFs,,,,, ---
```

designing-newsletters

```
---
name: designing-newsletters
description: Create branded newsletter layouts with the
templates and style guide in this
skill. Trigger when building or updating HTML email
newsletters.
```

skill-creator

```
---
name: skill-creator
description: Guide the creation of new agent skills:
draft SKILL.md, add references and
scripts, then package everything into a ZIP. Use/Trigger
when asked to build a skill.
```

Skill 3, 4, 5,,,,,,

Progressive disclosure – 2. instruction

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SKILL.md in pdf skill

PDF Processing Guide

Overview

This guide covers essential PDF processing operations using Python libraries and command-line tools. For advanced features, JavaScript libraries, and detailed examples, see REFERENCE.md. If you need to fill out a PDF form, read forms.md and follow its instructions.

Quick Start

```
```python
from pypdf import PdfReader, PdfWriter
Read a PDF
reader = PdfReader("document.pdf")
print(f"Pages: {len(reader.pages)}")
Extract text
text = ""
for page in reader.pages:
 text += page.extract_text()
...
```
```

Python Libraries

pypdf - Basic Operations

Merge PDFs

.....

Progressive disclosure – 3. resources

4 |

BE AN UNSTOPPABLE YOU

skills. Over time, you can become one of the best in your field if you keep it up. If your goal is to be the top salesperson in your company, be the most effective leader for your team, be a leading business owner in your industry, be a straight-A student at your college, or be the greatest athlete to play your game—effective self-leadership is the answer. As I reflected on self-leadership, a formula for how anyone can effectively lead themselves to peak performance began to take shape in my mind. This led to *Be an Unstoppable You: The Ultimate Self-Leadership Formula to Achieve PEAK Performance*, which took me more than four years to develop and allowed me to finally answer that one question I have pondered for so many years.

You Can Learn the Self-Leadership Formula™ to Achieve PEAK Performance

There are countless books on achieving peak performance lining people's bookshelves all over the world. Why should you read this one? Because it will work for you. Your ability to effectively lead yourself has a direct relationship to your ability to perform at the highest level. Each of us has the capability to have a peak-performance life. We may have started at different baselines, but we can rise above them. It does take some effort on your part. However, this effort will allow you to perform at your highest level and be unstoppable. PEAK performance is simply being the best you possible. *Be an Unstoppable You: The Ultimate Self-Leadership Formula to Achieve PEAK Performance* was developed to help anyone learn how to be their best. It provides the principles you need to achieve PEAK performance. Learning to effectively lead yourself is one of the most important things you will ever do. If you want to perform at your highest level and be unstoppable, it is necessary for you to learn the Self-Leadership Formula. In addition, it offers several other benefits:

YOU CAN ACHIEVE PEAK PERFORMANCE

| 5

BE AN UNSTOPPABLE YOU™

Provides a Clear Picture of How to Achieve PEAK Performance

How do people achieve PEAK performance? For those who are not familiar with the process, PEAK performance can appear unattainable. For them, achieving PEAK performance is reserved for people in a state of hypnosis; people who have special mental powers or people who are born with special ability. They may have a vague sense of what it takes for others to achieve PEAK performance, but they are unable to clearly see how their own effective self-leadership can make it possible for them. For others, self-leadership is a mystery, a confusing and overwhelming exercise. In contrast, *Be an Unstoppable You: The Ultimate Self-Leadership Formula to Achieve PEAK Performance* is visually straightforward, so anyone can learn it.

The smallest crowd you will
ever lead is you—but it is
the most important one.

BE AN UNSTOPPABLE YOU™

Defines the Principles of PEAK Performance

PEAK performance is based on principles, not myths or wishful thinking. In order to perform at your highest level and be unstoppable, it is important to understand these underlying principles. I wrote this book to help individuals like you recognize that effectively leading yourself requires you to hold true to the Principles of PEAK Performance.

The smallest crowd you will ever lead is you—but it is the most important one. Following the Principles of PEAK Performance is essential in advancing from one self-leadership formula variable to the next.

▼ pdf

▼ scripts

- check_bounding_boxes.py
- check_fillable_fields.py
- convert_pdf_to_images.py
- create_validation_image.py
- extract_form_field_info.py
- extract_form_structure.py
- fill_fillable_fields.py
- fill_pdf_form_with_annotations.py

forms.md

LICENSE.txt

reference.md

SKILL.md

forms.md in pdf skill

****CRITICAL: You MUST complete these steps in order. Do not skip ahead to writing code.****

If you need to fill out a PDF form, first check to see if the PDF has fillable form fields. Run this script from this file's directory:

``python scripts/check_fillable_fields <file.pdf>``, and depending on the result go to either the "Fillable fields" or "Non-fillable fields" and follow those instructions.

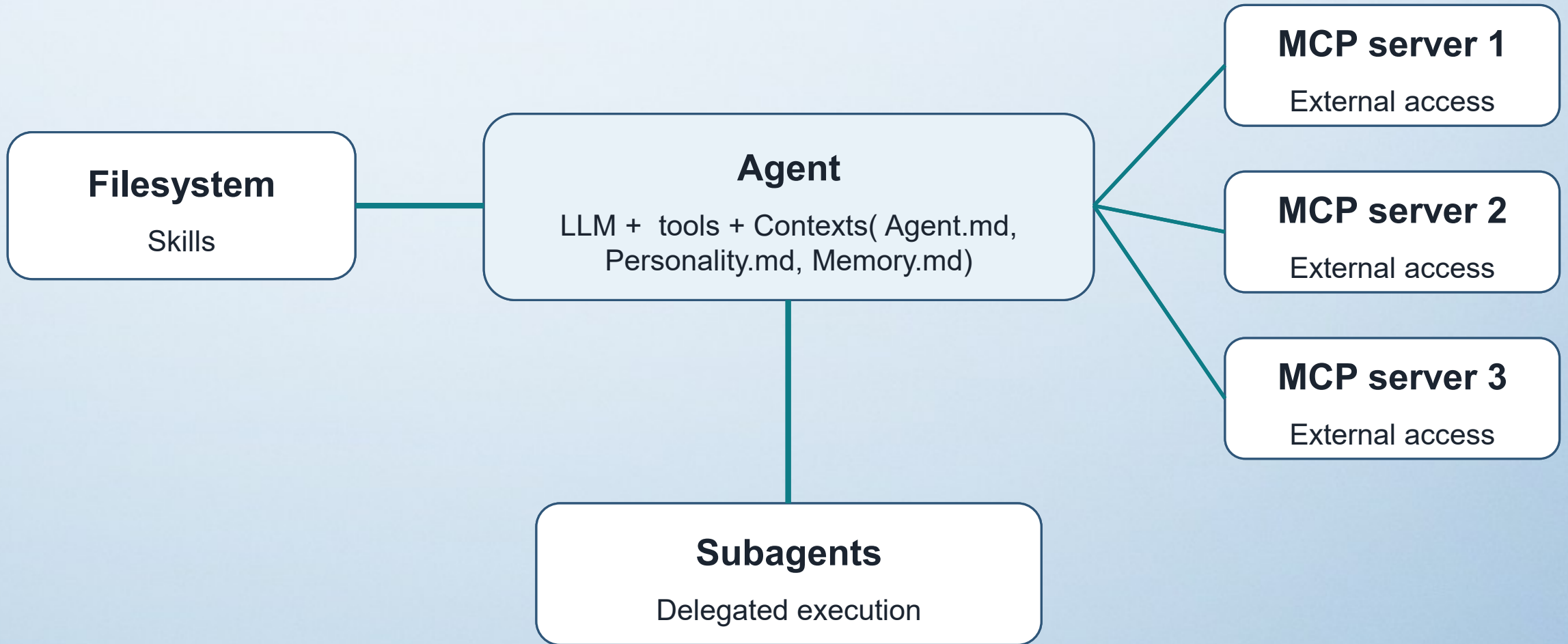
Fillable fields

.....

Skill in Agentic Ecosystem



Skills sit beside tools, MCP servers, and subagents



MCP provides access; Skills provide expertise

MCP

Connects the agent to databases,
APIs, and services.



Skills

Teach the agent what to do with the available
data.

Example: MCP retrieves a table; a skill explains how to compute metric X from columns A and B.

Tools stay in context; Skills arrive on demand

Tools

Essential capabilities

Read • Write • Edit • Bash
Glob • Grep • WebFetch • WebSearch
NotebookRead • NotebookWrite
Task • Skill

Definitions—name, description, and
parameters—always occupy context.

Skills

Specialized knowledge

Loads dynamically only when needed.

Can carry scripts that function as tools on
demand.

Extends what the agent knows how to do
without making every tool definition
permanent.

Subagents isolate work; Skills teach how to do it

Subagent

Own context
Own tool permissions
Works independently
Returns results

Skill

Provides expertise to the main agent or any subagent.

Example: language- or framework-specific review practices.

A Code Reviewer can be the delegated worker; a review skill provides the reusable standard.

Choose the mechanism that matches the need

| Mechanism | What it provides | Persistence | When it loads | Best for |
|-----------|-------------------------------|-----------------------|------------------|-----------------------|
| Skills | Procedural knowledge | Across conversations | Dynamically | Specialized expertise |
| Prompts | Moment-to-moment instructions | Single conversation | Each turn | Quick requests |
| Subagents | Task delegation | Across sessions | When invoked | Specialized tasks |
| MCP | Tool connectivity | Continuous connection | Always available | Data access |

Reference: <https://www.claude.com/blog/skills-explained>

Pre-built skills offer proven starting points

Anthropic's skills repo

github.com/anthropics/skills/tree/main/skills

pptx

Create and edit presentation decks.

skill-creator

Design a reusable skill package.

Start from a working pattern, then adapt it to your organization and workflow.

Building the Skill



SKILL.md has a small header and a powerful body

| | | |
|-------------|---|--|
| name | KL-patient-profile | |
| description | Generate a per-patient clinical profile markdown file from SDTM and/or ADaM .xpt datasets. Use this skill whenever the user wants to look at, summarize, review, or write up the data for an individual trial subject — demographics, disposition, adverse events, labs, vital signs, exposure, concomitant medications, medical history — from a study data folder. Trigger on requests like "make a patient profile", "what happened to subject 1015", "summarize this patient's AEs and labs", "profile the patients in this SDTM folder", "which patients are in this study", or any mention of USUBJID, SUBJID, ADaM, ADaE, ADLB, SDTM, or ADaM alongside a request about one subject. Also trigger when the user gives a study directory and a patient number without saying the word "profile". The skill requires the study data location as input up front and asks for the patient number if it wasn't supplied. Do NOT trigger for study-level or population-level analysis (summary tables, TLFs, incidence rates across arms) — this skill is about one patient at a time. | |

Patient profile from SDTM / ADaM

Produces a markdown profile for one subject, one `##` section per domain, by reading `.xpt` datasets straight from a study folder.

Paths below are relative to this skill's directory.

Input

Collect these from the user **before doing any processing**. Do not guess a study folder and do not profile a subject the user did not name.

| Input | Required | Notes |
|---------------------|--------------|---|
| Study data location | Yes | The study directory containing <code>.xpt</code> files. Any <code>sdm/</code> and <code>adam/</code> subfolders are found by walking the tree, so one path usually covers both. If the user gave separate SDTM and ADaM paths, pass their common parent or run once per folder. |
| Patient number | No — but ask | Bare <code>SUBJID</code> (<code>1015</code>) or full <code>USUBJID</code> (<code>01-701-1015</code>). If not given, do step 2 and ask which subject they want. |
| Domains | No | Comma list from <code>dm ds ex ae lb vs cm mh</code> ; omit for everything available. |
| Output file path | No | Defaults to <code>profile.md</code> in the working directory. |

If the study location is missing, ask for it and stop there — the rest of the process cannot start without it.

Output

1 YAML frontmatter

Metadata at the top tells an agent the skill name and when it applies.

2 Markdown body

Instructions define inputs, funnel metrics, efficiency metrics, output tables, and budget reallocation.

YAML frontmatter is a discovery contract

```
---  
name: KL-patient-profile  
description: Generate per-patient  
profile....  
---
```

N name

Maximum 64 characters

Lowercase letters, numbers, and hyphens only

No leading or trailing hyphens

Must match the parent directory

Gerund form is recommended

D description

Maximum 1,024 characters and non-empty

Describe what the skill does and when to use it

Include concrete keywords that improve discovery

**Good metadata helps the right
skill load at the right moment.**

Markdown body should guide action without becoming a manual

1

Recommended sections

Step-by-step instructions

Input and output formats

Examples

Common edge cases

2

Keep it lean

Stay under 500 lines

Show the basic path first

Move advanced detail to focused files

3

Keep it navigable

References one level deep

Clear, consistent terms

Forward slashes in every path

If a workflow grows large, move the detailed sequence into a separate file.

Optional folders separate execution, knowledge, and resources

/scripts

Executable helpers

Document dependencies
Explain whether Claude should execute or read
Return explicit, helpful errors

/references

Focused documentation

Read only when needed
Keep each file focused
Add a contents list above 100 lines

/assets

Reusable materials

Templates and configurations
Images, diagrams, and logos
Lookup tables and schemas

Skill Deployment, Invocation and Management



Import a marketplace in three visual steps

```
> /plug
/plugin (plugins)      Manage Claude Code plugins
```

1

Open plugin manager

Type /plugin and choose /plugins.

```
> /plugins

Add Marketplace

Enter marketplace source:
Examples:
  • owner/repo (GitHub)
  • git@github.com:owner/repo.git (SSH)
  • https://example.com/marketplace.json
  • ./path/to/marketplace

anthropics/skills
```

2

Add the marketplace

Select Add Marketplace and enter a source such as anthropics/skills.

```
> /skills

Skills
18 skills · Space to cycle, Enter to save, / to search, t to sort, Esc to cancel

Search skills...

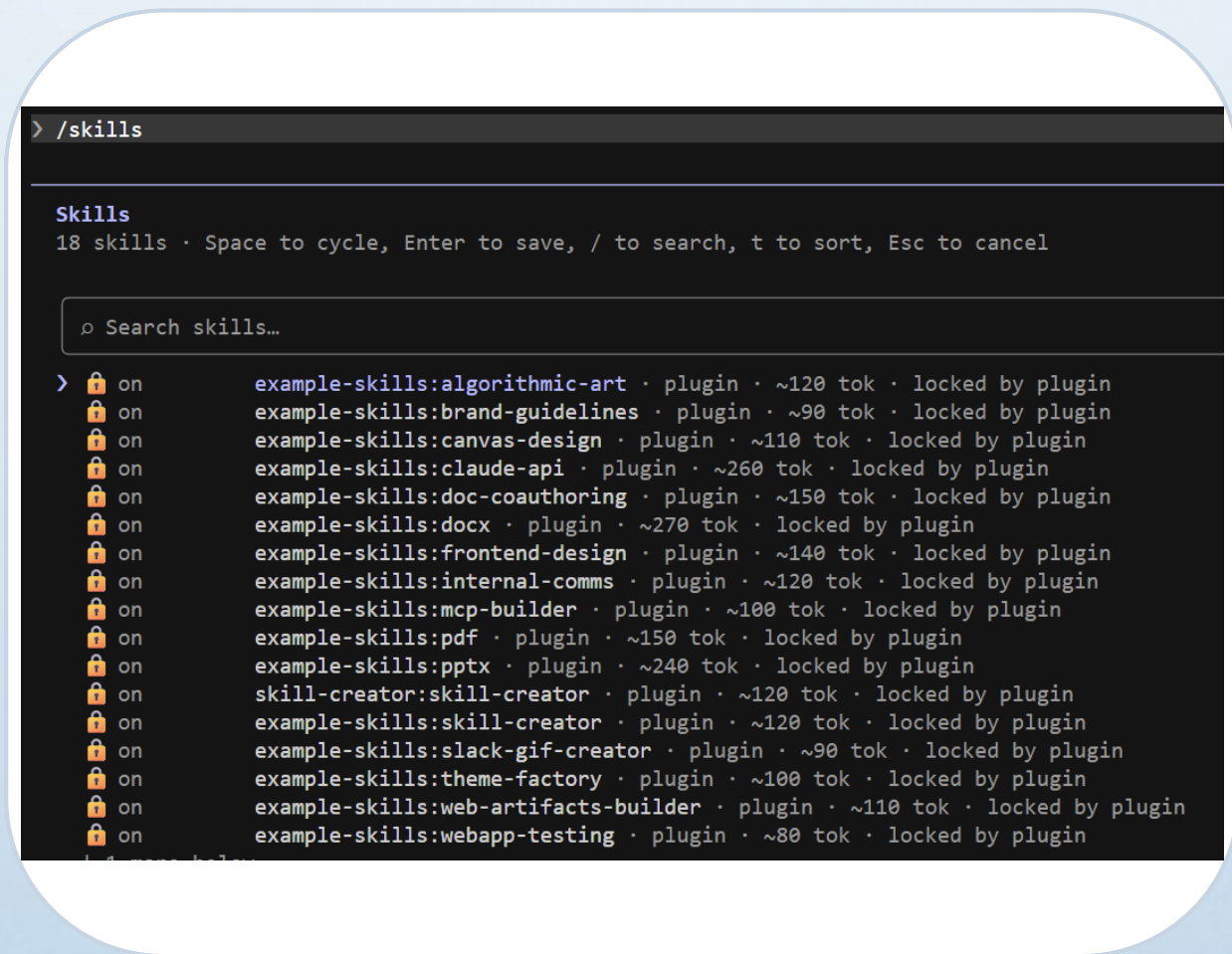
example-skills:algorithmic-art · plugin · ~120 tok · locked by plugin
example-skills:brand-guidelines · plugin · ~90 tok · locked by plugin
example-skills:canvas-design · plugin · ~110 tok · locked by plugin
example-skills:claude-api · plugin · ~260 tok · locked by plugin
example-skills:doc-coauthoring · plugin · ~150 tok · locked by plugin
example-skills:docx · plugin · ~270 tok · locked by plugin
example-skills:frontend-design · plugin · ~140 tok · locked by plugin
example-skills:internal-comms · plugin · ~120 tok · locked by plugin
example-skills:mcp-builder · plugin · ~100 tok · locked by plugin
example-skills:pdf · plugin · ~150 tok · locked by plugin
example-skills:pptx · plugin · ~240 tok · locked by plugin
skill-creator:skill-creator · plugin · ~120 tok · locked by plugin
example-skills:slack-gif-creator · plugin · ~90 tok · locked by plugin
example-skills:theme-factory · plugin · ~100 tok · locked by plugin
example-skills:web-artifacts-builder · plugin · ~110 tok · locked by plugin
example-skills:webapp-testing · plugin · ~80 tok · locked by plugin
```

3

Browse and toggle skills

Search, inspect token size, and turn capabilities on or off.

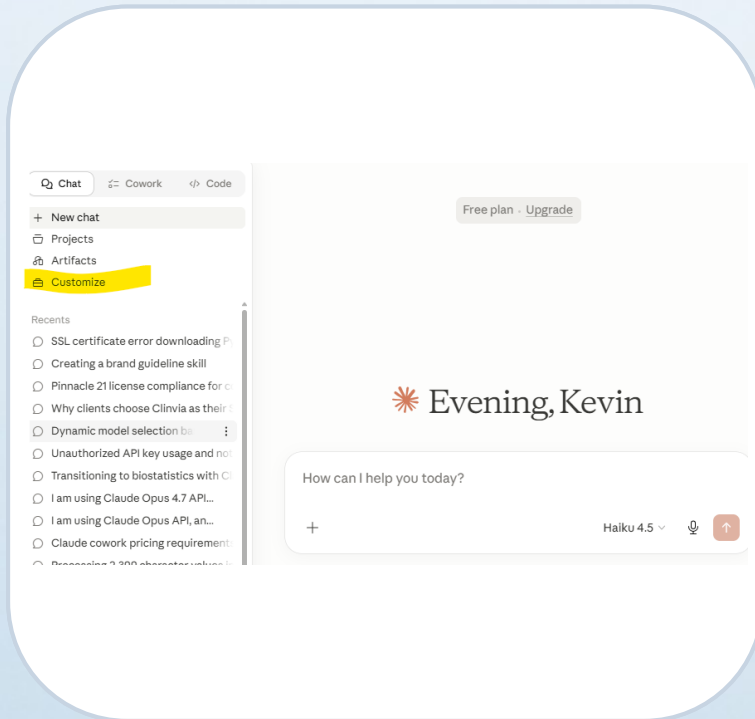
The skills browser makes the capability catalog visible



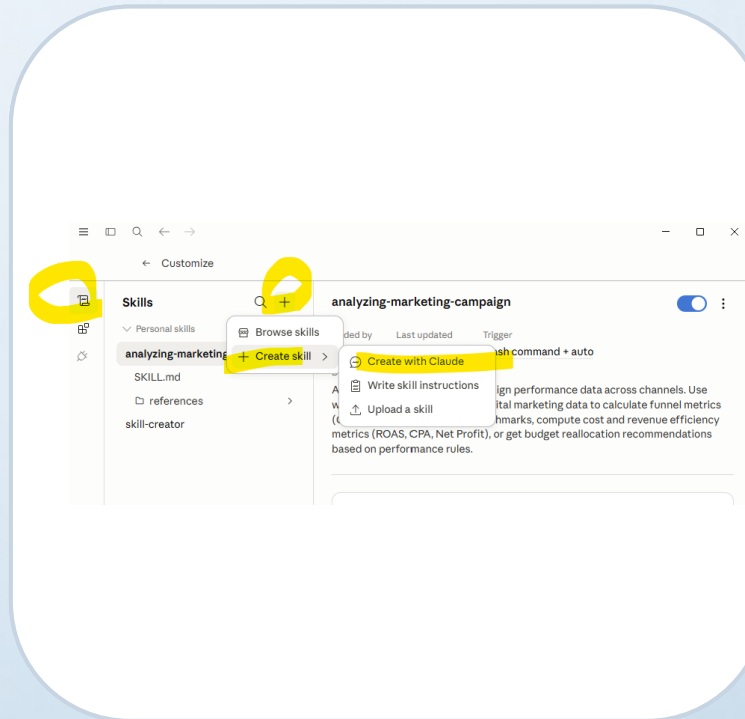
| | |
|------------------|----------------|
| algorithmic-art | internal-comms |
| brand-guidelines | mcp-builder |
| canvas-design | pdf |
| claude-api | pptx |
| doc-coauthoring | skill-creator |
| docx | theme-factory |
| frontend-design | webapp-testing |

Searchable • Toggleable • Token-aware

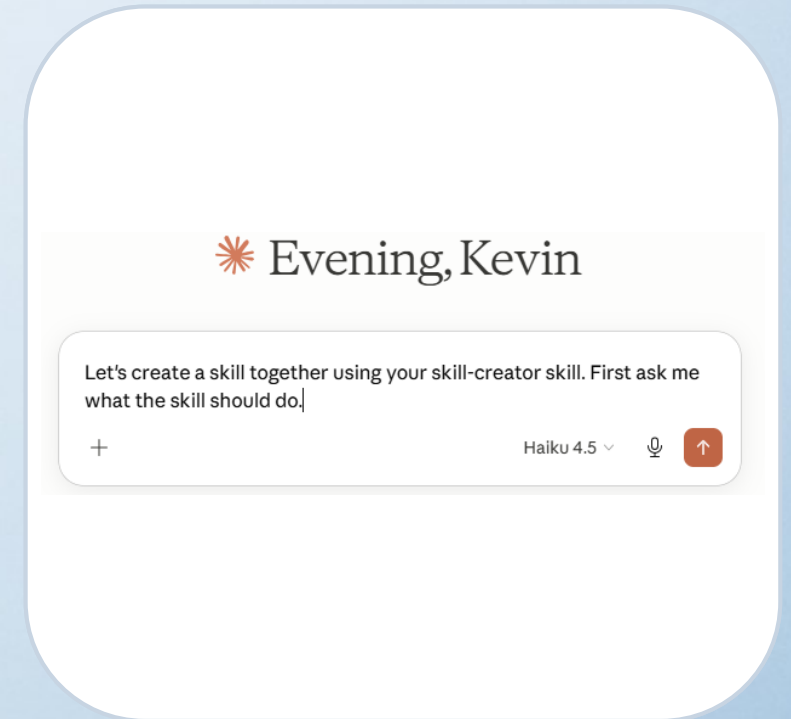
Create a skill with Claude.ai from the Customize panel



1 Open Customize



2 Choose Create with Claude



3 Describe the skill together

Alternatives: write skill instructions yourself or upload an existing skill.

Hands-on Workshop



